

Session 1: Introduction to R

RStudio



& don't forget to create an **R project** for each new analysis!

1. Script: write, run & save code
to run code: *command/control + enter*
2. Console: see results, run quick code
3. Environment: variables
4. Files/plots/help

Comments

Comments start with a **hashtag** & are ignored by R. Use them to add notes to your code *# just like this! :)*

Maths

addition	+	division	/
subtraction	-	exponents	^
multiplication	*		

Brackets `()` can be used to specify order of operations, e.g. `(1 + 3) * 2`

Comparisons

equal to	==
not equal to	!=
less/greater than	> or <
less/greater or equal to	<= or >=

Give logical (**TRUE/FALSE**) as output

Variables

Use `<-` to assign, e.g. `x <- 10`

Naming variables:

- **no** spaces
- **no** special characters (e.g. ! \$ &)
- **no** starting with number
- **avoid** capital letters
- **avoid** names that aren't descriptive

Functions

function name
↓
`round(3.1459, digits = 2)`
↑
surrounded by brackets

comma separates arguments

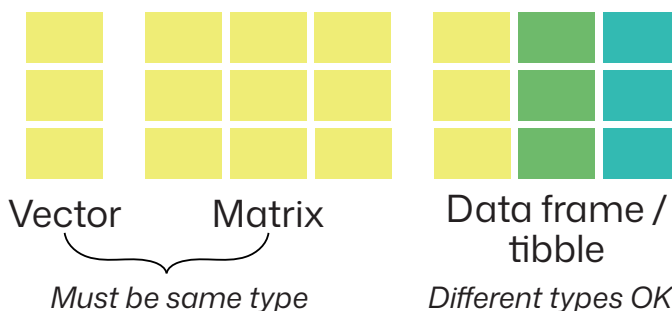
Get help with `?function_name`

Atomic data types

logical (true/false)	TRUE
numeric (numbers)	1000
character (text)	"hello"

Check type with the `class()` function

Data structures



Summary sheet

Create vectors with `c()` e.g. `c(1, 2, 3)`

Create data frames from a series of vectors with `data.frame()`

e.g. `data.frame(`
 `row1 = c(1, 2),`
 `row2 = c("a", "b"))`

See just the start with `head()`, just the end with `tail()` or the open the whole thing with `View()`

R packages

Packages are collections of functions, e.g. the tidyverse. **Install once** using `install.packages("tidyverse")` and **load every time you open R** using `library("tidyverse")`

Paths

folder names file name
↓ ↓
`"my_analysis/data/genes.csv"`
↑ ↑
/ to separate file extension

Reading files

assign data to variable

↓
`genes <- read_delim(`
 `"my_analysis/data/genes.csv")`
↑
file path

Use `read_delim()` for any file, or specifically `read_csv()/read_tsv()`